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L&T Assignment
Configuration of Address Resolution Protocol (ARP) using Cisco Packet Tracer

Aim

The aim of this project is to construct a simple Local Area Network (LAN) and study the operation of the Address Resolution Protocol (ARP) using Cisco Packet Tracer.
The project demonstrates how devices in a network communicate using IP addresses and MAC addresses, and how ARP resolves an IP address to its corresponding MAC address to enable communication between devices in the same network.

This experiment helps in understanding the relationship between Layer 2 and Layer 3 addressing in computer networks and shows how address resolution is required before data transmission can occur in Ethernet networks.

GitHub link: https://github.com/Adarshbhau/ARP-LAN-Configuration

Introduction

Computer networks rely on unique addressing mechanisms to allow devices to communicate with each other. Two major addressing schemes used in networking are IP addresses
and MAC addresses. An IP address is a logical address assigned to devices to identify them within a network. A MAC address is a physical hardware address assigned to the network interface card (NIC) of a device.

While IP addresses are used for routing packets across networks, the actual transmission of frames inside a local network occurs using MAC addresses. Because of this difference, there must be a mechanism that translates IP addresses into MAC addresses so that frames can be delivered correctly. This mechanism is provided by the Address Resolution Protocol (ARP).

In this project, a simple LAN is constructed using PCs and an Ethernet switch in Cisco Packet Tracer. The behavior of ARP requests and ARP replies is analyzed to understand how devices dynamically discover the MAC addresses of other devices in the network before sending data packets.

What is an IP Address

An Internet Protocol (IP) address is a unique logical identifier assigned to every device connected to a computer network that uses the Internet Protocol for communication. The IP address allows devices to locate each other and exchange information over the network.

IP addresses are used at the Network Layer (Layer 3) of the OSI model and are essential for routing packets between networks. Routers analyze IP addresses in packet headers to determine the best path to forward the data.

IPv4 addresses consist of 32 bits and are usually represented in dotted decimal format such as 192.168.1.10. Each IP address is divided into four octets separated by dots, and each octet can have a value between 0 and 255.

IPv6 addresses were introduced to overcome the limitations of IPv4 address space. IPv6 uses 128-bit addressing and is represented in hexadecimal format separated by colons.

IP Address Classes

IPv4 addresses were traditionally divided into five classes: A, B, C, D, and E.

Class A addresses are used for very large networks and have the first octet ranging from 1 to 126.
Class B addresses are designed for medium-sized networks and range from 128 to 191 in the first octet.
Class C addresses are commonly used in small networks and range from 192 to 223.
Class D addresses are reserved for multicast communication.
Class E addresses are reserved for experimental purposes.

Although classful addressing has largely been replaced by CIDR (Classless Inter-Domain Routing), the concept of IP address classes is still useful for understanding how addressing evolved in networking.

What is a MAC Address

A Media Access Control (MAC) address is a unique physical address assigned to a network interface card by the manufacturer. Unlike IP addresses, MAC addresses are generally
permanent and stored in the hardware of the network device.

A MAC address consists of 48 bits and is typically written in hexadecimal format such as 00:1A:2B:3C:4D:5E.

The first 24 bits represent the Organizationally Unique Identifier (OUI), which identifies the manufacturer of the network interface. The remaining 24 bits uniquely identify the device produced by that manufacturer.

MAC addresses operate at the Data Link Layer (Layer 2) of the OSI model and are used to deliver frames within a local area network.

Address Resolution Protocol (ARP)

Address Resolution Protocol (ARP) is used to map an IP address to a MAC address within a local network. When a device wants to send data to another device on the same network,
it must know the MAC address of the destination device.

If the MAC address is not already known, the sending device broadcasts an ARP request asking which device owns the specific IP address. The device with the matching IP address
responds with an ARP reply that contains its MAC address.

Once the MAC address is received, the sending device stores it in its ARP cache table and uses it for future communication until the entry expires.

ARP therefore acts as the bridge between the Network Layer and the Data Link Layer in Ethernet networks.

Reverse Address Resolution Protocol (RARP)

Reverse Address Resolution Protocol (RARP) performs the opposite function of ARP. Instead of translating an IP address into a MAC address, RARP translates a MAC address into
an IP address.

RARP was historically used by diskless workstations that did not have storage to store their IP configuration. When such a system started, it only knew its MAC address. It would broadcast a RARP request to a RARP server asking for its assigned IP address.

The RARP server maintained a table of MAC addresses and corresponding IP addresses. When it received the request, it would respond with the appropriate IP address.

However, RARP had limitations. It required a dedicated server and could only provide IP addresses without additional configuration parameters. Because of these limitations, RARP has largely been replaced by BOOTP and DHCP.

Problem Statement

In modern computer networks, communication between devices depends on both logical addressing and physical addressing. While IP addresses are used for identifying devices across networks, actual communication within a local network requires MAC addresses.

When a device wants to send data to another device using its IP address, it must first determine the corresponding MAC address of the destination device. Without a protocol to perform this translation automatically, network communication within a LAN would not be possible.

Therefore, a mechanism is required to dynamically map IP addresses to MAC addresses so that devices can communicate efficiently within the network.

Scope of the Solution

The scope of this project is to design and simulate a simple LAN network using Cisco Packet Tracer and analyze how the Address Resolution Protocol enables communication between devices.

The project focuses on understanding network addressing, observing ARP request and reply messages, analyzing ARP tables in devices, and verifying communication using ping
commands.

Through simulation, students gain practical knowledge of networking concepts such as Ethernet communication, switching, addressing, and protocol behavior.

Required Components

Hardware Components:
Personal Computers (PCs)
8-Port Ethernet Switch
LAN Cables (Ethernet Cables)
Network Interface Cards

Software Components:
Cisco Packet Tracer for network simulation
Operating System with command prompt for running networking commands
Optional IDE tools for configuration and documentation

These components are sufficient to construct and analyze a basic LAN environment and observe ARP operations.

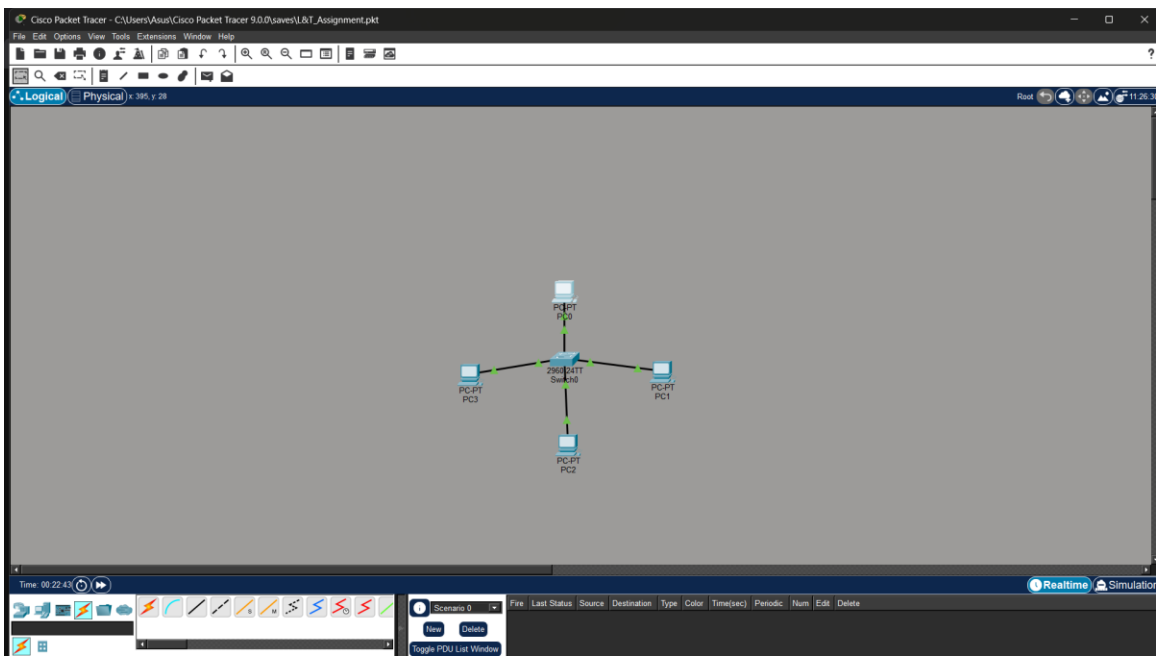
Simulated Circuit and Procedure

The simulated network topology consists of two PCs connected to an 8-port Ethernet switch using copper straight-through cables.

Step 1: Open Cisco Packet Tracer.
Step 2: Drag and place two PCs from the end devices section.
Step 3: Place an Ethernet switch in the workspace.
Step 4: Connect the PCs to the switch using copper straight-through cables.
Step 5: Assign IP addresses to both PCs.
Example:
PC1 – 192.168.1.1
PC2 – 192.168.1.2
Subnet Mask – 255.255.255.0

Step 6: Open the command prompt on PC1 and run the ping command to test connectivity.
Step 7: Observe ARP request and ARP reply packets in simulation mode.
Step 8: Use the command 'arp -a' to display the ARP table.

Screen Shots/Outputs:



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PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

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Cisco Packet Tracer PC Command Line 1.0
C:\>pin 192.168.1.2\
Invalid Command.

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.2           0060.3e73.83c0        dynamic

C:\>
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
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C:\>
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Top

Conclusion

This project successfully demonstrated the working of the Address Resolution Protocol in a Local Area Network using Cisco Packet Tracer. The experiment showed how devices resolve IP addresses into MAC addresses before transmitting data frames.

By observing ARP request and reply messages, the project helped illustrate the interaction between network layer addressing and data link layer addressing. The simulation
provided practical insight into how Ethernet communication functions and highlighted the importance of ARP in ensuring successful network communication.